Adrin Alias
Engineering Portfolio

linkedin.com/in/adrinalias
University of Texas at Arlington
B.S. Mechanical Engineering, December 2025

Brake Dyno

Sensors & Data Acquisition

August 2024 – May 2025

Firmware Development	Circuit Design
Signal Conditioning	CAN Communication
Python	SolidWorks (CAD)

What & Why

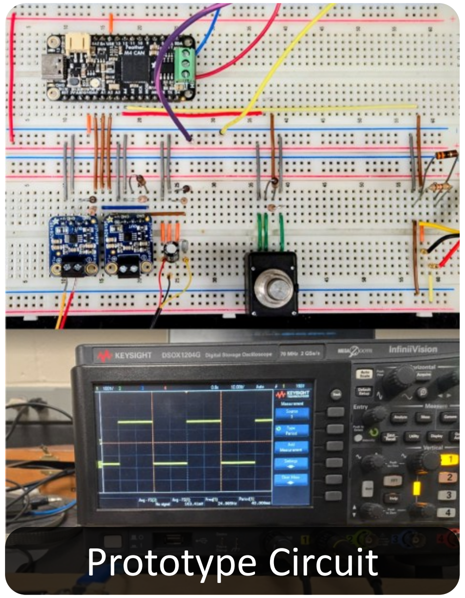
Developed a Brake Dynamometer (6-person team) for the Formula SAE team to measure the coefficient of friction for various brake components, enabling data-driven design decisions to improve lap times. I was responsible for the complete development of its custom DAQ system, including the hardware, firmware and GUI software.

How

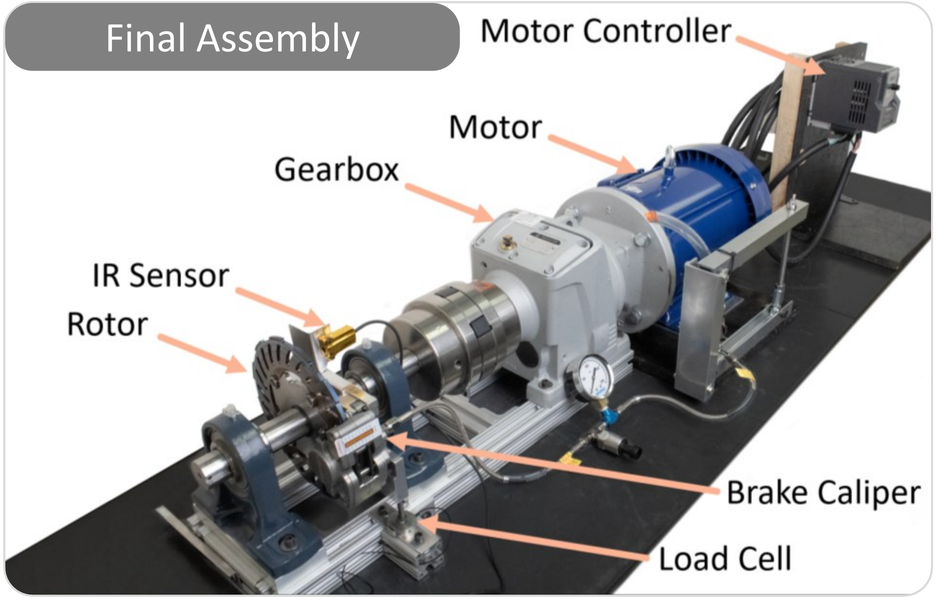
The test rig uses a motor to spin a brake assembly while measuring torque, speed, and temperature. I developed the DAQ system using an SAME51 MCU, which involved designing the **signal conditioning circuits** (voltage dividers, RC filters) and programming the **C++ firmware** to process all sensor data, manage timing, and transmit it via Serial. I also created a **Python GUI** for real-time data plotting & logging.

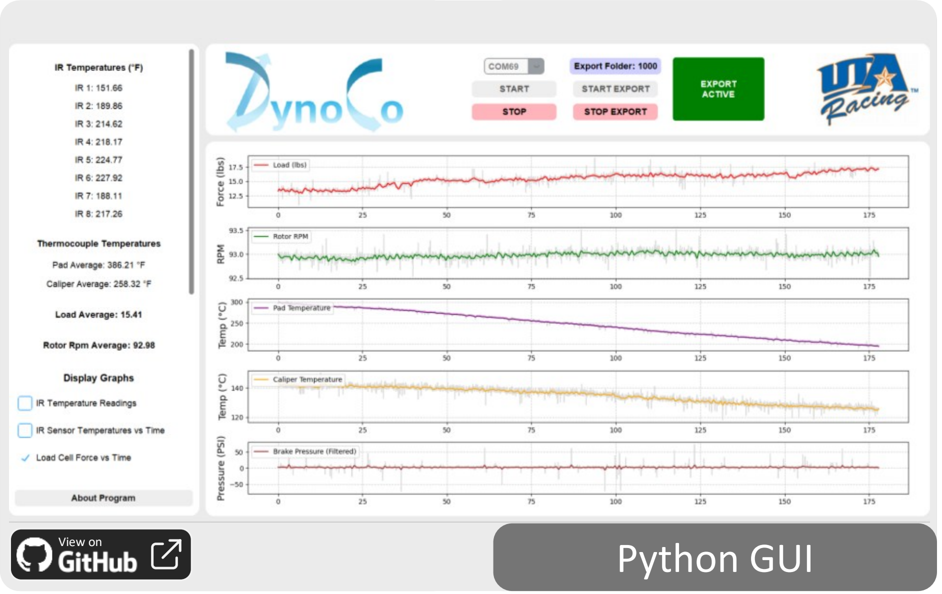
Results

- Delivered a functional DAQ with **20 Hz** real-time processing and logging, producing a coefficient of friction vs. temperature graph that **validated manufacturer claims.**
- Reduced system cost by **94%** (from $8200 to $500) compared to a commercial NI DAQ by focusing on core requirements.
- Awarded 2nd Place at UTA Innovation Day 2025.

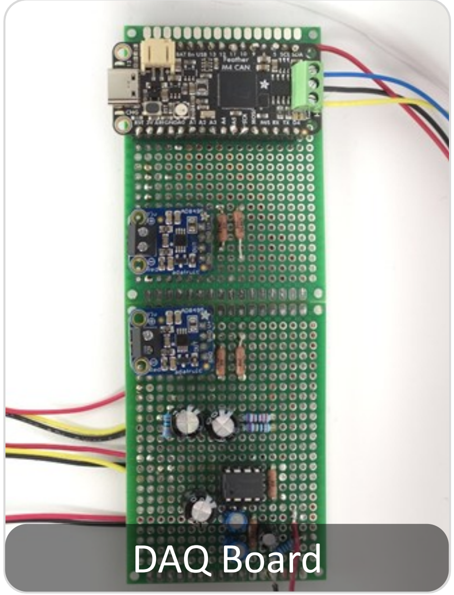

Prototype Circuit

Final Assembly





Python GUI


DAQ Board

IoT Smart Binds

Powered by ESP32

May 2024 – August 2024

- SolidWorks (CAD)
- 3D Printing
- Enclosure Design
- Rapid Prototyping
- Firmware Development
- Testing & Validation

What & Why

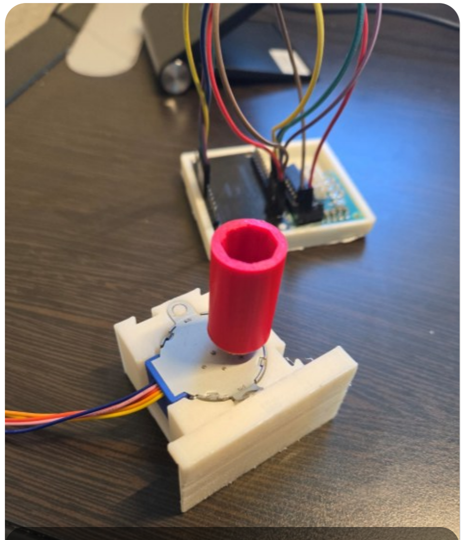
- A small electromechanical device to automate opening & closing blinds at a lower cost than commercial alternatives.
- Personal project to improve sleep quality and quantity by only letting in light from wake-up time till dusk.

How

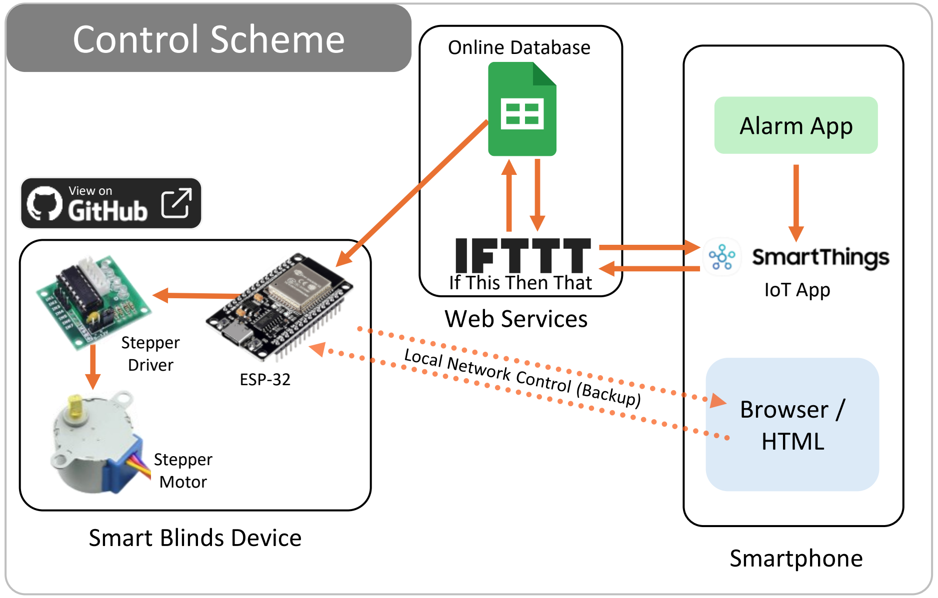
- Utilized an ESP32 microcontroller and an off-the-shelf stepper motor driver. Programmed the device in C++ (Arduino) for automation and open-loop motor control.
- The firmware connects via Wi-Fi, fetches daily sunset times from a public weather API and syncs with an NTP server for timekeeping. Samsung SmartThings is used to detect the morning alarm.

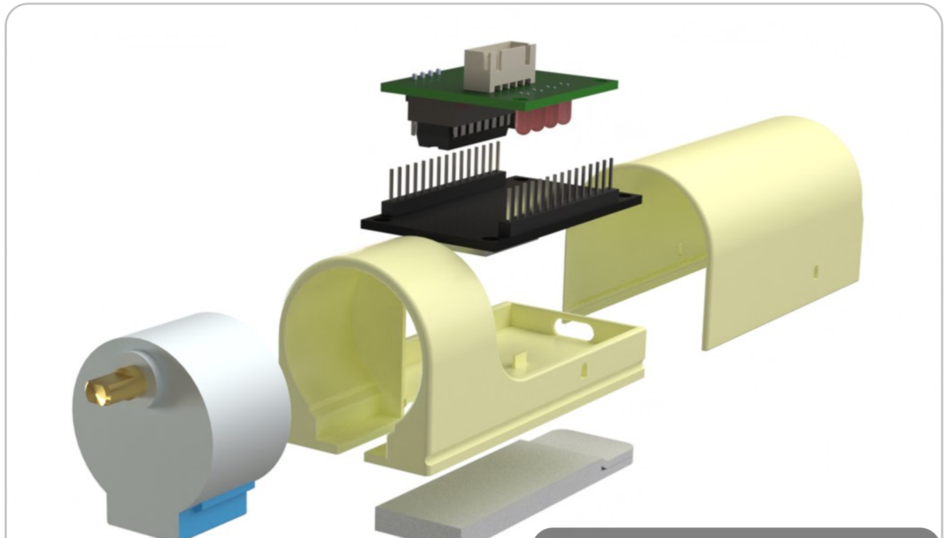
Results

- Successfully created a **fully autonomous system** that reliably opens and closes daily based on user alarm info and daily sunset times.
- **85% lower hardware cost** than similar alternatives in the market. ($70 → $10)
- Completed a 30-day continuous testing with **100% reliability** in opening and closing.
- *Future Work:* Upgrading system to use an H-bridge and a car power window motor to open bigger blinds with strings.



1st Prototype





Simplified Assembly

Final Version

Air Heaters

Enclosures & Control Systems

SolidWorks (CAD)	Manufacturing
Digital Twin	Sheet Metal Design
Electrical Wiring	Circuit Design

What & Why

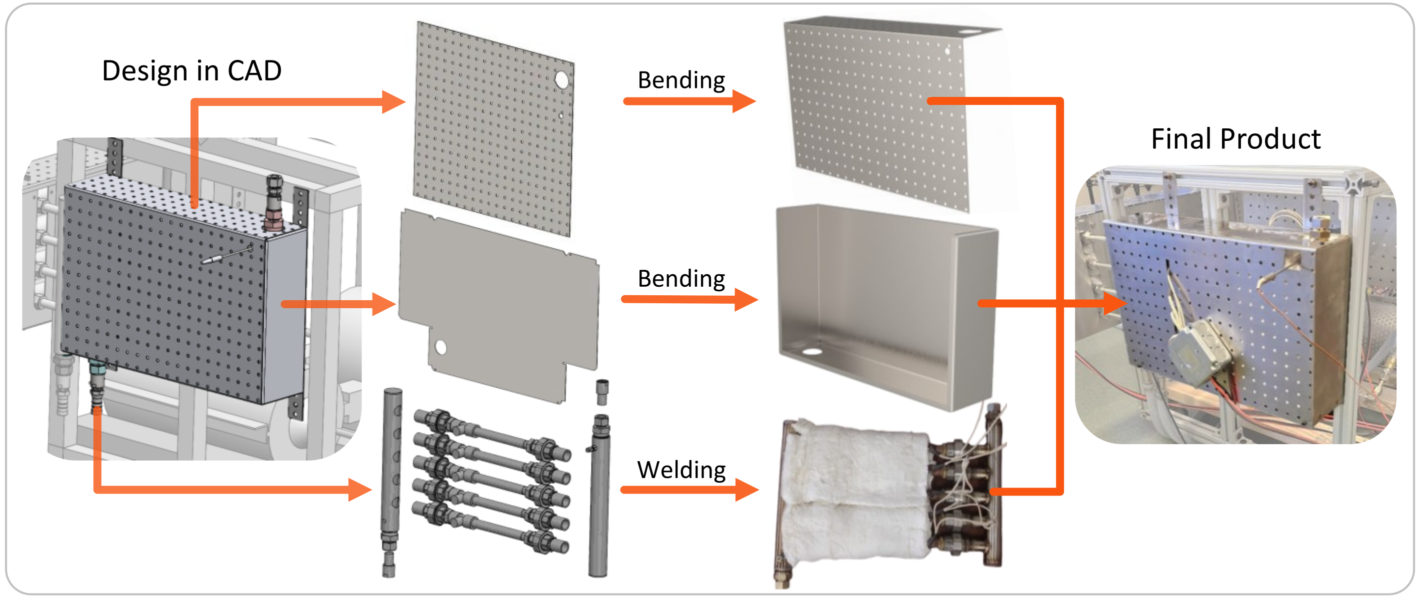
- Part of a **high temperature test rig** to simulate the thermodynamic state of a jet engine turbine that needs temperatures up to **750 °C**, net air flowrate of **85 SCFM** at pressures up to 60 psi.
- Required a custom solution because there was no off-the-shelf solution. (Price info unavailable)
- I was responsible for the design of the heater enclosures & the design & integration of the power & control systems.

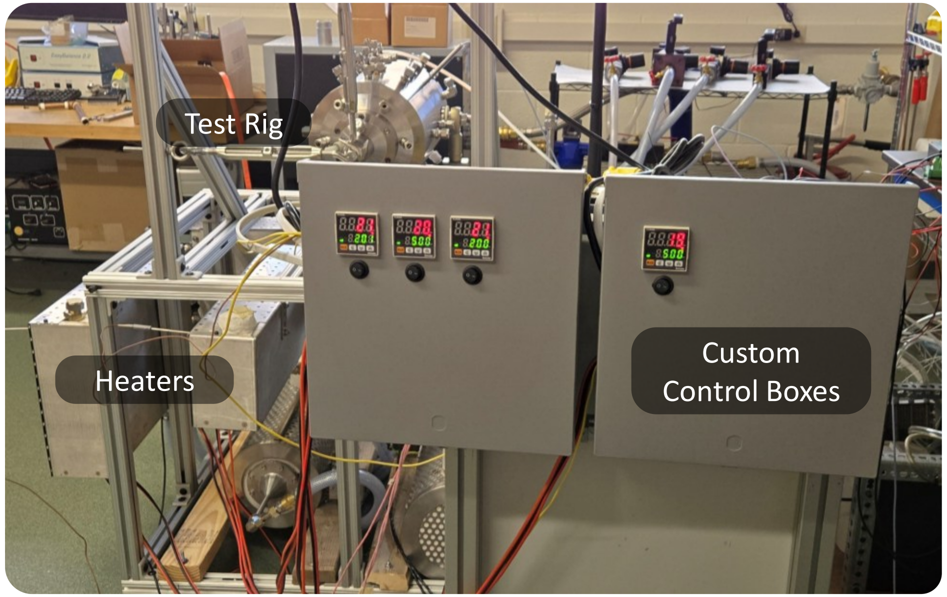
How

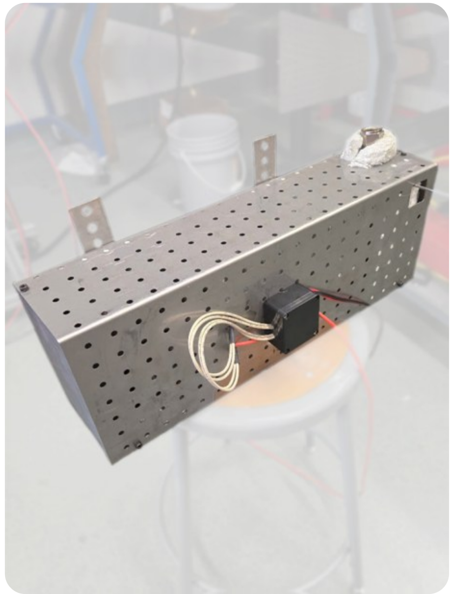
- Designed the SS 304 enclosures in SolidWorks, applying DFMA principles to streamline production with external fabrication shops.
- Collaborated with the university machine shop for manufacturing and welding.
- Personally assembled, tested, and wired all four heater modules, integrating them with an off-the-shelf PID controller to create a complete, functional system.

Results

- Successfully implemented an **8.4 kW air heater system** across 4 heaters and achieved the target temperatures and flow rates.
- **Reduced manufacturing cost 30%** by outsourcing sheet metal fabrication.







Turbine Blade

Generation & Modelling

Research Assistant, TESLAB | March 2024 – July 2024

MATLAB	CAD
Parametric Modelling	FEA

What & Why

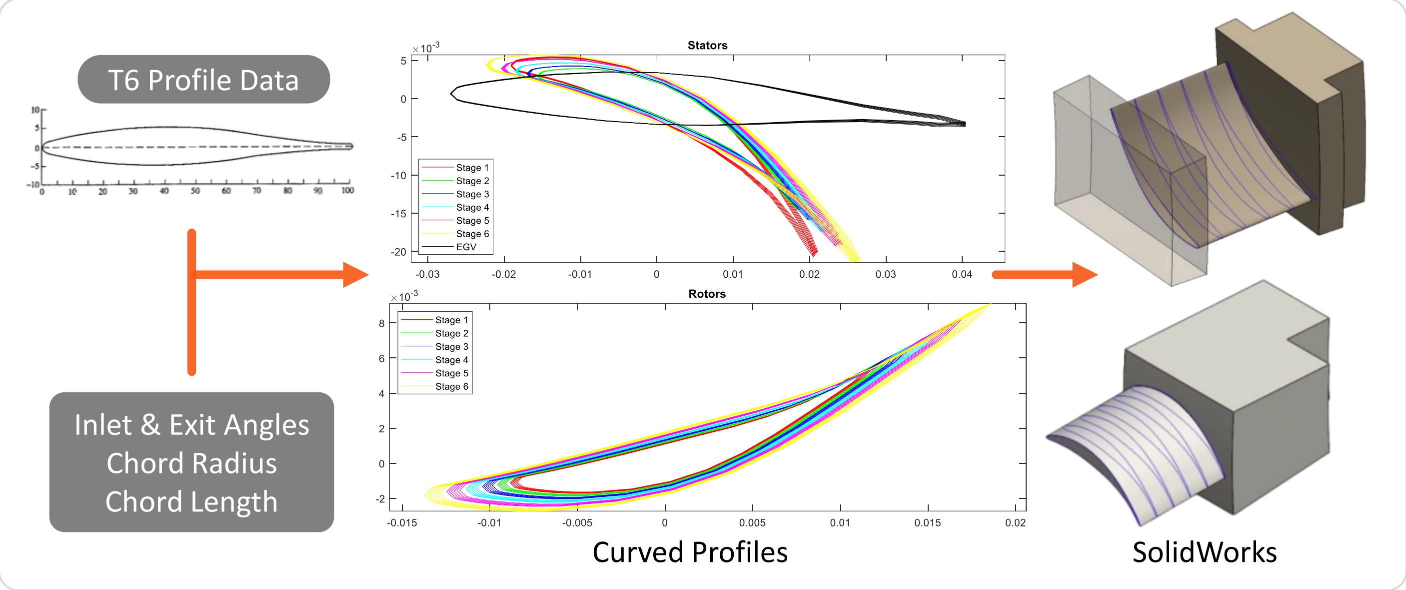
- Transformed a T6 turbine airfoil cross-section into curved airfoils for turbine rotors & stators.
- Blade designs for a multistage Helium Turbine.

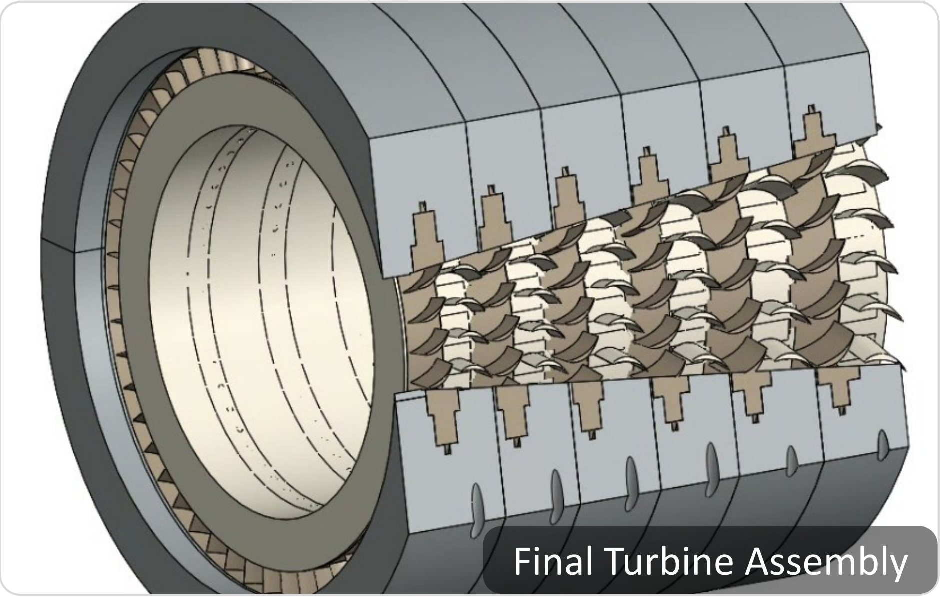
How

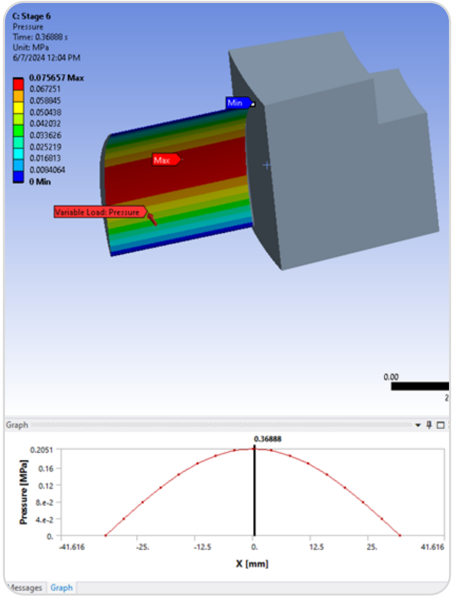
- Used previously calculated entry & exit angles, and curvature radii to bend the T6 profile in MATLAB.
- Imported profiles as points into SolidWorks & modelled each rotor & stator.
- Developed a macro to speed up importing workflow.
- Conducted preliminary FEA in Ansys to check for stress concentrations.

Results

- Developed easily modifiable code to bend any turbine blade profiles.
- Created complete assembly of turbine and compressor assembly in SolidWorks for the proposal submission.







Trebuchet

Design Project

March 2022 – May 2022

CAD	Laser Cutting
GD&T	3D Printing
Rapid Prototyping	DFMA
Testing	Simulation

What & Why

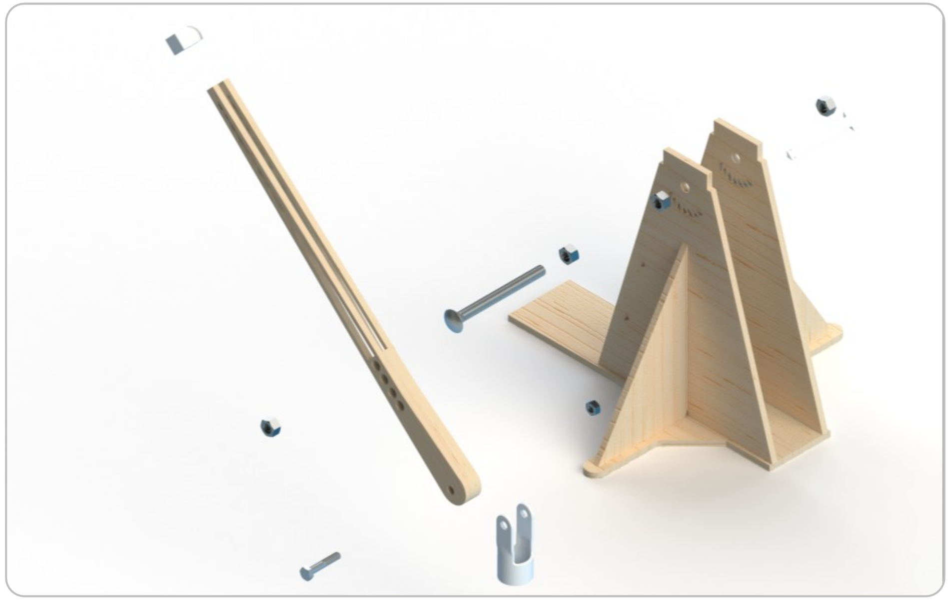
- Design and build a trebuchet using computer-controlled manufacturing methods to launch a penny using quarters exactly 6 feet.

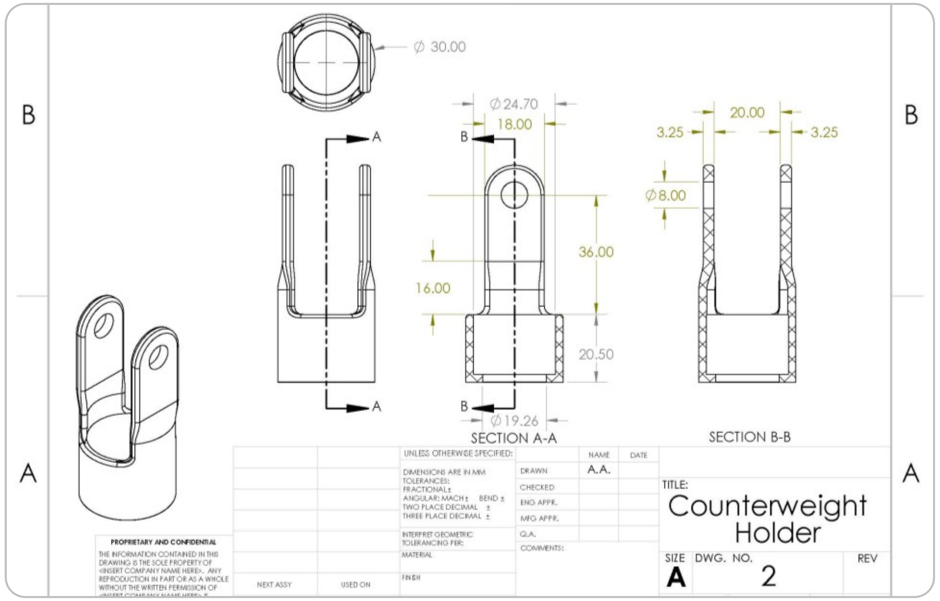
How

- Designed in SolidWorks to be manufactured using laser cut wood, 3D printing & COTS components only.
- Laser cut parts assembled only with interference fits due to tolerancing studies.
- Used initial prototype to test and correct simulation data.

Results

- Trebuchet was able to launch the penny as designed and did not break any design requirements.





Mobius Racing

Formula 1 in Schools Competition

December 2020 – January 2021

CAD	First Principles
Fixture Design	Design of Experiment
Rapid Prototyping	3D Printing
DFMA	CFD

What & Why

- Design, manufacture & race a miniature racecars powered by a CO_2 canister (F1 in Schools Competition).
- Team represented high school in National Finals.
- My Roles: Team Lead, Vehicle Manufacturing & Assembly, & Wheel Design.

How

- Used Fusion360 to model different configurations of the wheel and conduct elementary FEA.
- Conducted DOEs to controls variable for 3D printing to maximize layer adhesion & part strength.
- Design fixtures to align the parts of the car during assembly.
- Conducted tests to check dimensional accuracy to verify interference levels for the bearings.

Results

- Fastest car in the competition.
- Selected to represent the nation in the World Finals.
- Judge's pick for Best Engineered Car.

1st Prototype

Tolerancing Tests